The Unruh Effect as Kullback–Leibler Divergence

A Natural Information-Geometric Formulation with a Calculable Mode-Dependent Correction

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Abstract

We show that the Unruh effect admits an exact formulation in terms of Kullback–Leibler (KL) divergence between observer-dependent partial descriptions of the quantum vacuum. The Minkowski vacuum is identified as the complete description G, and the Rindler thermal state — established by the Bisognano–Wichmann theorem [3] independently of any specific calculation — as the partial description P(o_R) obtained by tracing over the causally inaccessible left wedge. The KL divergence between the two Gaussian descriptions has the **exact closed form**, valid for all frequencies and accelerations:

$$D_{KL}(P(o_R) \| P(o_M)) = 2\bar{n}(\omega,a) - \ln(2\bar{n}(\omega,a) + 1)$$

where $\bar{n} = 1/(e^{2\pi\omega/a} - 1)$ is the Planck distribution. This result is **not** merely the Unruh temperature restated: it is a mode-resolved information quantity that reduces to $D_{KL} \propto T_U/\omega$ only in the classical limit $\omega/a \to 0$, and **falls faster than $\bar{n}$** in the intermediate regime $\omega/a \sim 0.1$–1, with deviations of 66–100% from the classical approximation.

The principal results are: (1) the Unruh effect has a natural and exact information-geometric formulation; (2) the exact KL expression $D_{KL} = 2\bar{n} - \ln(2\bar{n}+1)$ constitutes a mode-dependent information quantity that differs substantially from the standard Planck distribution in the quantum regime; (3) this difference constitutes a calculable, falsifiable prediction: the information content per mode falls faster than the particle number per mode for $\omega/a \gtrsim 0.1$. We discuss how this prediction could in principle be tested in acoustic analogue experiments.

Scope. This paper reformulates, not re-derives, the Unruh effect. The thermal character of ρ_R is established by Bisognano–Wichmann independently. The novelty is in the exact mode-resolved KL expression and its deviation from the classical approximation.

Keywords: Unruh effect, KL divergence, information geometry, Rindler spacetime, Bisognano–Wichmann, acoustic analogues, mode-resolved decoherence.

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1. Introduction

The Unruh effect [1] states that a uniformly accelerated observer with proper acceleration a perceives the Minkowski vacuum as a thermal bath at temperature

$$T_U = \hbar a / (2\pi c k_B)$$

This result is usually stated as a single temperature — a uniform property of the perceived vacuum. The mode-resolved version, the Planck distribution $\bar{n}(\omega) = 1/(e^{2\pi\omega/a}-1)$, gives the mean occupation number per mode but does not distinguish the information content of the observation from its particle content.

The information-geometric framework of [2] provides a natural setting to ask this question. In that framework, the Rindler observer's partial description $P(o_R)$ is a submanifold of the complete description G (the Minkowski vacuum), obtained by tracing over the causally inaccessible degrees of freedom. The information gap $\Delta(o_R) = D_{KL}(G \parallel P(o_R))$ measures the cost of this inaccessibility. The question becomes: what is the exact mode-resolved form of this information gap?

The answer, derived in Section 4, is:

$$D_{KL}(\omega, a) = 2\bar{n}(\omega,a) - \ln(2\bar{n}(\omega,a) + 1) \quad \text{[exact, all } \omega \text{ and } a\text{]}$$

This differs from $\bar{n}(\omega)$ in a precise, calculable way. In the classical limit $\omega/a \to 0$, $D_{KL} \approx 2\bar{n}$ and scales with T_U. But for $\omega/a \gtrsim 0.1$, D_{KL} falls faster than $\bar{n}$ by a logarithmic correction that reaches 66–100% suppression in the intermediate quantum regime. This is the new, calculable result of this paper.

> **Central question of this paper:** if information and particle number are different quantities ($D_{KL} \neq \bar{n}$), which one controls the physical observables in an analogue system? We do not answer this definitively — it requires a dynamical theory of how D_{KL} enters the coupling — but we quantify the difference precisely and identify the experimental regime where it matters.

2. Background

2.1 The Bisognano–Wichmann Theorem

The Bisognano–Wichmann theorem [3], proved for general Wightman field theories, establishes that the restriction of the vacuum state to the right Rindler wedge is a KMS thermal state at inverse temperature $\beta = 2\pi/a$:

$$\rho_R = \mathrm{Tr}_L(|0_M\rangle\langle 0_M|) = Z^{-1} \exp(-(2\pi/a) H_R)$$

This is more general than the Unruh (1976) derivation via Bogoliubov coefficients: it applies to all Wightman theories without specifying particle content. The temperature $T_U = a/(2\pi)$ follows as a consequence, not an assumption.

> We use Bisognano–Wichmann as the input that establishes the thermal character of ρ_R. The KL divergence computed in Section 4 therefore does not assume the Unruh temperature; it recovers its natural scale from the structure of the Gaussian states.

2.2 Information-Geometric Framework

In the framework of [2], a physical system is described by a complete statistical manifold G satisfying statistical sufficiency (C1), reparametrisation invariance (C2), and locality (C3). A partial description $P(o)$ is a submanifold obtained by marginalising over inaccessible degrees of freedom. The information gap $\Delta(o) = \inf_{q\in P(o)} D_{KL}(p_G \parallel q)$ drives the emergent temporal parameter τ via accumulated updates.

The identification is exact: $P(o_R) \equiv \rho_R$ (the marginalisation Tr_L is the definition of $P(o_R)$), and $G \equiv |0_M\rangle\langle 0_M|$. The Rindler horizon is the information boundary $\partial P(o_R)$.

3. Why $D_{KL} \neq \bar{n}$ Is Non-Trivial

The standard characterisation of the Unruh effect gives $\bar{n}(\omega)$ as the thermal occupation number per mode. The quantity $D_{KL}(\omega,a)$ computed here is fundamentally different:

Quantity	**$\bar{n}(\omega,a)$**	**$D_{KL}(\omega,a)$**
Physical meaning	Mean particle number (Fock space)	Information gap (statistical manifold)
Classical limit ($\omega/a \to 0$)	$\approx a/(2\pi\omega)$	$\approx a/(\pi\omega) - \ln(a/(\pi\omega))$
Quantum limit ($\omega/a \to \infty$)	$\to 0$ exponentially	$\to 0$ exponentially (faster)
Relation	—	$D_{KL} = 2\bar{n} - \ln(2\bar{n}+1) < 2\bar{n}$
Geometric role	Fock space occupation	KL divergence on statistical manifold

The correction term $-\ln(2\bar{n}+1)$ is always negative and grows in relative importance as ω/a increases. This is not a small perturbative correction: at $\omega/a = 0.2$, D_{KL} is **87% smaller** than $2\bar{n}$; at $\omega/a = 0.5$, it is **99% smaller**. The two quantities agree only in the deep classical regime.

The physical interpretation: $\bar{n}$ counts quanta of energy; D_{KL} counts bits of information lost to the horizon. These are related but distinct. At low frequencies (long wavelengths), a photon carries little information per quantum of energy, so the two scale together. At high frequencies, a quantum carries more information, and the logarithmic correction captures the saturation of the information content per mode.

4. Exact Calculation

4.1 Gaussian States

For a free massless scalar field, both partial descriptions are Gaussian. For a single mode ω, in quadrature representation:

$P(o_M)$: covariance $\Sigma_M = (1/2)\, I_2$ [vacuum, zero mean]

$P(o_R)$: covariance $\Sigma_R = (\bar{n} + 1/2)\, I_2$ [thermal, zero mean]

4.2 Exact Closed Form

The KL divergence between two zero-mean Gaussians with scalar covariances σ_R^2 and σ_M^2 in n=2 dimensions:

$$D_{KL} = (1/2)[\, tr(\Sigma_M^{-1}\Sigma_R) - 2 + \ln(\det \Sigma_M / \det \Sigma_R)\,]$$

$$= \sigma_R^2/\sigma_M^2 - 1 - \ln(\sigma_R^2/\sigma_M^2)$$

$$= (2\bar{n} + 1) - 1 - \ln(2\bar{n} + 1)$$

$$= 2\bar{n} - \ln(2\bar{n} + 1) \quad \text{[exact for all } \omega/a\text{]}$$

4.3 Numerical Results

Table 1. Exact KL divergence vs. classical approximation $2\bar{n}$, as a function of $x = \omega/a$. The normalized correction $C(x) = (\pi \cdot x) \cdot D_{KL}$ measures the deviation from the Unruh temperature scale ($C=1$ in classical limit).

$x = \omega/a$	$\bar{n}(x)$	D_{KL} (exact)	$C(x)$ [norm.]	Dev. from T_U	Regime
0.01	15.42	27.38	0.860	14%	≈classical
0.05	2.71	3.56	0.559	44%	transition
0.10	1.14	1.10	0.345	66%	transition
0.20	0.40	0.21	0.132	87%	quantum
0.50	0.045	0.0039	0.006	99%	quantum
1.00	0.0019	7×10^{-6}	~0	~100%	deep quantum

$C(x) = (\pi \cdot x) \cdot D_{KL}(x)$: normalized so that $C=1$ in the classical limit $\omega/a \to 0$. Values less than 1 indicate suppression relative to the Unruh temperature scale.

4.4 Limiting Behaviour

Classical limit ($\omega/a \to 0$):

$$\bar{n} \to a/(2\pi\omega) \gg 1$$

$$D_{KL} \approx 2\bar{n} - \ln(2\bar{n}) = a/(\pi\omega) - \ln(a/(\pi\omega))$$

Leading term: $D_{KL} \propto T_U / \omega$ [Unruh temperature recovered]

Quantum regime ($\omega/a \sim 0.1 - 1$, the physically relevant transition):

$$D_{KL} = 2\bar{n} - \ln(2\bar{n}+1) \ll 2\bar{n}$$

Suppression: $D_{KL} / (2\bar{n}) = 1 - \ln(2\bar{n}+1)/(2\bar{n}) \to 0$

The logarithmic correction $-\ln(2\bar{n}+1)$ accounts for 66–100% of the information content in this regime. This is where D_{KL} and $\bar{n}$ are maximally different and where any information-geometric effect would be most visible.

5. Interpretation

5.1 D_{KL} as the Cost of a Horizon

The KL divergence $D_{KL}(\omega,a)$ is the information cost, per mode, of having a Rindler horizon. It measures how much information is lost when the Minkowski vacuum is restricted to the right wedge. In the framework of [2], this is the per-mode contribution to $\Delta(o_R) = D_{KL}(G \parallel P(o_R))$.

The fact that $D_{KL} < 2\bar{n}$ (not $= 2\bar{n}$) reflects a saturation effect: once a mode is in a highly thermal state (large $\bar{n}$), the information gap grows slower than the particle number because the state is already maximally mixed in the relevant subspace. The logarithmic correction captures this saturation.

5.2 Relation to Entropy

The von Neumann entropy of the Rindler thermal state per mode is:

$$S_R(\omega) = (\bar{n}+1)\ln(\bar{n}+1) - \bar{n}\ln(\bar{n}) \quad \text{[von Neumann entropy]}$$

In the classical limit, $S_R \approx \ln(\bar{n}) + 1$ while $D_{KL} \approx 2\bar{n}$. These are very different quantities. D_{KL} measures the divergence from the vacuum; S_R measures the intrinsic entropy of the thermal state. The Bekenstein–Hawking entropy formula $S_{BH} = A/(4G\hbar)$ is related to the integrated S_R, not directly to D_{KL}.

> Open question: is D_{KL} or S_R the more fundamental quantity for horizon thermodynamics? The two differ by $\ln(2\bar{n}+1)$ per mode. Answering this requires a dynamical theory of how information gaps enter physical couplings.

5.3 What This Reformulation Does Not Establish

Stating explicitly what remains open:

- Whether the decoherence rate of a system coupled to the Rindler vacuum is proportional to D_{KL} or to $\bar{n}$. The standard QFT answer is $\bar{n}$. For D_{KL} to be the relevant quantity, the coupling would need to be information-geometric rather than Hamiltonian — an assumption not proved here.
- Whether the exact expression $D_{KL} = 2\bar{n} - \ln(2\bar{n}+1)$ can be derived without assuming the Bisognano–Wichmann thermal character. This would require showing that the variational principle $S[P(o)]$ forces the KMS condition, which is an open problem.
- Whether the logarithmic correction has any observable consequence in current experiments.

6. Experimental Implications

6.1 Acoustic Analogue Systems

The most advanced experimental platforms for Unruh/Hawking physics are acoustic analogues in Bose–Einstein condensates (BEC). Steinhauer (2016) [5] and subsequent experiments have measured the analogue Hawking radiation spectrum in the regime $\omega/a_{eff} \sim 0.1$–2, exactly where D_{KL} and $\bar{n}$ differ substantially.

In these systems, the analogue acceleration $a_{eff} = c_s^2/\xi$ (where c_s is sound speed and ξ is the healing length) sets the analogue Unruh temperature. The measured spectrum is the mode occupation $\bar{n}_{meas}(\omega)$, compared against the Planck distribution at T_U.

A measurement of D_{KL} in these systems would require measuring not the particle number but the KL divergence between the output state and the vacuum state — a quantum tomography measurement. This is technically feasible in BEC systems where full state tomography is achievable for low-dimensional mode spaces.

6.2 The Measurable Prediction

The prediction is: the information-geometric distance between the thermal Rindler state and the Minkowski vacuum, measured via KL divergence on the Gaussian state space, should follow:

$$D_{KL,meas}(\omega) = 2\bar{n}_{meas}(\omega) - \ln(2\bar{n}_{meas}(\omega) + 1)$$

rather than a simple proportionality $D_{KL} \propto \bar{n}$ or $D_{KL} \propto T_U/\omega$. In the regime $\omega/a \sim 0.2$–0.5, this predicts a **87–99% suppression** of the information content relative to the naive classical estimate.

This is **falsifiable in principle**: if the measured KL divergence follows $2\bar{n} - \ln(2\bar{n}+1)$, the information-geometric interpretation is confirmed. If it follows $\bar{n}$ or $2\bar{n}$, the Hamiltonian coupling dominates and the information-geometric correction is not physically relevant. The experiment distinguishes between these by measuring the full covariance matrix of the output state (quantum tomography), not just the mean photon number.

> Practical caveat: current BEC experiments do not routinely perform full quantum tomography on the output modes. The prediction is falsifiable with next-generation experiments where state reconstruction is feasible. We flag this as a target for experimental collaborators.

7. Discussion

7.1 What Is Genuinely New

To be explicit about the novelty:

- (1) The exact expression $D_{KL} = 2\bar{n} - \ln(2\bar{n}+1)$ is, to our knowledge, not present in the existing literature on information-theoretic approaches to the Unruh effect. Most treatments compute the von Neumann entropy or the mutual information, not the KL divergence between the two observer states.
- (2) The mode-resolved information gap $D_{KL}(\omega,a)$ is a fundamentally different quantity from both $\bar{n}(\omega)$ and $S_R(\omega)$. The difference is calculable, large in the intermediate regime, and in principle observable.
- (3) The identification of the Rindler observer as a partial description $P(o_R)$ in the sense of [2] is precise and complete, providing a dictionary between horizon thermodynamics and information geometry.

7.2 What Remains Open

- Whether D_{KL} or $\bar{n}$ controls the physical decoherence rate in a coupled system. This is the central dynamical question.
- Whether the KMS condition can be derived from $S[P(o)]$ without Bisognano–Wichmann as input.
- Whether the correction $-\ln(2\bar{n}+1)$ appears in the Bekenstein–Hawking formula for black holes.
- Full quantum tomography in BEC analogue experiments to test the prediction of Section 6.

7.3 Relation to Existing Information-Theoretic Approaches

Several authors have applied information theory to the Unruh and Hawking effects. Adesso et al. [6] computed the quantum mutual information between Rindler modes. Fuentes-Schuller and Mann [7] studied entanglement between inertial and accelerated observers. The present approach differs in computing the KL divergence between the complete description and the partial description, rather than between subsystems of the partial description. The two quantities are related but distinct.

8. Conclusions

We have derived the exact, mode-resolved KL divergence between the Minkowski vacuum and the Rindler thermal state:

$$D_{KL}(\omega, a) = 2\bar{n}(\omega,a) - \ln(2\bar{n}(\omega,a) + 1) \quad \text{[exact, all } \omega \text{ and } a\text{]}$$

This expression:

- Reduces to $D_{KL} \approx 2\bar{n}$ in the classical limit $\omega/a \to 0$, recovering the Unruh temperature scale.
- Deviates from $2\bar{n}$ by 66–100% in the physically accessible intermediate regime $\omega/a \sim 0.1$–1. This deviation is the logarithmic correction $-\ln(2\bar{n}+1)$.
- Is in principle testable in acoustic analogue experiments via quantum tomography of the output state.

The paper is a reformulation of the Unruh effect in information-geometric language. Its value lies not in replacing the standard derivation but in identifying D_{KL} as the natural information-geometric quantity associated with a horizon, computing it exactly, and quantifying where it differs from the standard particle-number description. The open problem — whether this difference is physically observable — is the agenda for future work.

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